DETAILED DESCRIPTION OF THE INVENTION

Data Structures and Encoded Geometry Representation

In various embodiments, the reconstructed penile anatomical geometry model may be converted into a structured digital data representation configured for secure storage, transmission, and computational interoperability. The geometry profile may comprise a multidimensional dataset including longitudinal position indices, radial distance values, cross-sectional parameters, curvature vectors, volumetric estimations, and derived proportionality metrics.

The geometry representation may be stored as a parametric vector set, a mesh-based data structure, a spline-defined surface model, or other computationally efficient encoding format. In certain embodiments, the system may reduce raw surface data into a compressed descriptor comprising normalized feature vectors that preserve structural characteristics while minimizing data size. Such encoding may exclude raw imagery while retaining sufficient parameters to recreate or approximate the three-dimensional structural model.

In some implementations, the geometry profile may include hierarchical data layers. A first layer may contain raw or filtered radial sampling values indexed by longitudinal displacement. A second layer may contain reconstructed cross-sectional parameters. A third layer may contain derived aggregate metrics such as taper ratios, symmetry indices, and curvature classifications. A fourth layer may include normalized comparison scores relative to stored reference distributions.

The encoded geometry representation may be formatted according to standardized schema definitions enabling interoperability with external calibration systems, compatibility modeling frameworks, personalization engines, or encrypted cloud storage infrastructures. Metadata fields may include timestamp information, device identification codes, measurement confidence metrics, physiological validation flags, and cryptographic authentication markers.

In certain embodiments, a geometry profile may be transformed into an abstracted structural signature comprising a condensed set of encoded parameters suitable for anonymized comparison or matching algorithms. Such structural signatures may enable compatibility scoring or clustering without exposing explicit anatomical dimensions.

The data architecture described herein is not limited to any specific encoding format, compression method, schema definition, or storage structure, provided that the representation preserves sufficient spatial information to describe a user-specific penile anatomical geometry profile derived from synchronized time-of-flight distance and motion measurements.